

SHRUTI JETHVA

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EDUCATION

New Jersey Institute of Technology — B.S. Biomedical Engineering

Newark, NJ | Expected May 2027

- Relevant Coursework: Biomechanics, Human Physiology, Biomedical Instrumentation, Circuits, Engineering Design

WORK EXPERIENCE

RVL Pharmacy — Pharmacy Technician (Part-Time)

RVL Pharmacy | [City, State] | 2023 – Present

- Processed and dispensed 100+ specialty prescriptions daily in a high-volume private pharmacy, maintaining accuracy and regulatory compliance across all fulfillment workflows
- Guided patients through out-of-pocket reimbursement processes with their insurance providers, improving medication accessibility and reducing financial barriers for specialty drug patients at a cash-pay pharmacy
- Managed compounding and handling of specialty medications requiring precise preparation protocols, gaining direct exposure to pharmaceutical standards and drug safety procedures
- Maintained medication inventory and controlled substance records across an active specialty formulary, ensuring stock accuracy and regulatory traceability

PROJECTS

Arduino-Based Dialysis Monitoring System

FED101: Fundamentals of Engineering Design | Spring 2025

- Coordinated a 4-person team to design and build a functional dialysis simulation device using Arduino Uno, a TDS sensor, DC motor pump, servo motor, and LCD display
- Programmed sensor-to-display logic in C++ to read solute concentration in mg/L and trigger color-coded LED alerts when TDS readings exceeded clinically-defined filtration thresholds
- Fabricated a 3D-printed kidney shell and laser-cut anatomical patient model to simulate a catheterized dialysis setup, demonstrating cross-disciplinary applicability in medical and pharmaceutical monitoring

Machine Learning for Gait Disorder Classification & Symmetry-Index Prediction

Individual Project | 2025

- Built and compared 4 ML classifiers (Random Forest, KNN, SVM-RBF, Gradient Boosting) on 8,000 walking trials from the GaitRec database to classify lower-limb gait pathology across 4 diagnostic categories
- Achieved 62.8% classification accuracy with SVM-RBF — approximately 2.5x above random baseline — using engineered biomechanical features including bilateral peak force ratios and gait symmetry index
- Developed a Gradient Boosting regressor to predict Gait Symmetry Index with $R^2 = 0.909$ and RMSE = 2.05%, within the 3% clinical engineering target threshold
- Processed and feature-engineered 75,000+ force-plate trials across 101 time-normalized stance-phase samples using Python and scikit-learn

EMG-Controlled Tank Robot

BME383: Biomedical Instrumentation | 2025

- Assembled a wireless muscle-controlled robot integrating Biopac MP36 EMG and hand dynamometer signals through an LM324 analog signal-conditioning circuit
- Built full signal chain from electrode acquisition through ESP-NOW wireless transmission to dual DC motor control via DRV8833 H-bridge driver on an ESP32 microcontroller
- Performed hands-on soldering, circuit assembly, and hardware troubleshooting; isolated a receiver firmware-to-chip communication fault, preserving the functional transmitter and wireless link

RESEARCH EXPERIENCE

Undergraduate Research Assistant

Biomechanics Research Lab, NJIT | Spring 2025

- Contributed to lab meetings and project development in a biomechanics research environment; assisted with experimental setup, equipment configuration using Biopac hardware, and literature review
- Gained hands-on exposure to biomechanical data collection protocols and lab instrumentation standards

LANGUAGES & TECHNOLOGIES

Licensure: New Jersey State Pharmacy Technician License

Programming: Python, MATLAB, C++

Software: SolidWorks, Biopac (AcqKnowledge/BSL), Microsoft Excel

Hardware & Fabrication: Arduino, ESP32, breadboard circuit assembly, soldering, 3D printing, laser cutting

Techniques: EMG signal processing, ground reaction force analysis, machine learning (scikit-learn), TDS sensing